

Lab Report 02

1/ The square root of the sum of squares of all elements in the matrix.

```
#include <stdio.h>

#include <math.h>

int main()
{
    int size;

    printf("Enter the matrix size: ");
    scanf("%d",&size);

    int arr[size][size];
    for(int i=0; i<size; i++){
        for(int k=0; k<size; k++){
            scanf("%d",&arr[i][k]);
        }
    }

    //The square root of the sum of squares of all elements in the matrix:
    int sum;
    for(int i=0; i<size; i++){
        for(int k=0; k<size; k++){
            sum += arr[i][k]*arr[i][k];
        }
    }

    double normal= sqrt(sum);
    printf("The normal of the matrix: %.2f\n", normal);
    return 0;
}
```

2/ Trace of matrix to daiagonal elements:

```
#include <stdio.h>

int main() {

    int size;

    printf("Enter the matrix size: ");
    scanf("%d", &size);


    int matrix[size][size];

    printf("Enter the elements: \n");
    for (int i = 0; i < size; i++) {
        for (int k = 0; k < size; k++) {
            scanf("%d", &matrix[i][k]);
        }
    }


    int trace = 0;
    for (int i = 0; i < size; i++) {
        trace += matrix[i][i];
    }

    printf("The trace of the matrix: %d\n", trace);
    return 0;
}
```

3/ Check if a given square matrix is an upper triangular matrix:

```
#include <stdio.h>

int main(){

    int size;

    printf("Enter the matrix size: ");

    scanf("%d",&size);

    int matrix[size][size];

    printf("Enter the element: \n");

    for(int i=0; i<size; i++){

        for(int k=0; k<size; k++){

            scanf("%d",&matrix[i][k]);

        }

    }

    //Check if a given square matrix is an upper triangular matrix:

    int sum=0;

    for(int i=1; i<size; i++){

        for(int k=0; k<size-2; k++){

            sum += matrix[i][k];

        }

    }

    if(sum==0){

        printf("The given matrix is upper triangular matrix.\n");

    }

    else{

        printf("The given matrix isn't upper triangular matrix.");

    }

    return 0;

}
```

4/ Find the length of a character

```
#include <stdio.h>

int main(){

    char str_one[50];

    printf("Enter the first name: ");

    scanf("%s", str_one);


    char str_two[50];

    printf("Enter the second name: ");

    scanf("%s", str_two);


    int length = 0;

    for(int i = 0; str_one[i] != '\0'; i++){

        length++;

    }


    int length2 = 0;

    for(int i = 0; str_two[i] != '\0'; i++){

        length2++;

    }

    printf("The length of string is : %d\n", length);

    printf("The length of string is: %d\n", length2);

}
```

5/ To compare the two string:

```
#include<stdio.h>
```

```
#include<string.h>
```

```
int main(){
```

```
    char a[50];
```

```
    printf("Enter the first name: ");
```

```
    scanf("%s",&a);
```

```
    char b[50];
```

```
    printf("Enter the seceond name: ");
```

```
    scanf("%s",&b);
```

```
    // Compare two string:
```

```
    if(strcmp(a,b) ==0){
```

```
        printf("String are equal.\n");
```

```
    }
```

```
    else{
```

```
        printf("String are not equal.");
```

```
    }
```

```
    return 0;
```

```
}
```